

Laboratory Disorganization and the Potential of Digital Chemistry Learning in Jaffna Schools

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Introduction & Research Background

- **Role of Practical Work:**
  - ✓ Essential for G.C.E. Advanced Level Chemistry.
  - ✓ Builds analytical skills, safety habits, and STEM competence.
- **Observed Local Challenges:**
  - ✓ Expired chemical stockpiles (no standardized disposal mechanism).
  - ✓ Structural gaps: poor ventilation, limited fume hoods, missing safety pits.
  - ✓ Risk of environmental contamination (groundwater/soil) and health issues.
- **The Paradigm Shift:**
  - ✓ Inquiry learning loses impact when experiments are omitted or memorized.
  - ✓ Need to examine digital alternatives to overcome physical limitations.

Objectives & Research Questions

- **Primary Objective:**
  - ✓Assess perceptions of laboratory disorganization, safety, and environmental impact
  - ✓ while evaluating digital learning potential.
- **Research Questions:**
  1. *Disorganization:* What forms of physical/structural disorganization exist?
  2. *Impact:* How do laboratory conditions associate with safety and environment?
  3. *Digital Solutions:* How can digital models enhance safe, inquiry-based chemistry?

Methodology & Study Design

- **Study Design:** Cross-sectional, perception-based survey.
- **Target Area:** 16 1AB Schools in Valikamam Educational Zone, Jaffna.
- **Sample Size (N):**
  - ✓**Teachers:** 15 Chemistry Educators (Complete Zone Census).
  - ✓**Students:** 244 G.C.E. A/L Science Students (Krejcie & Morgan Sampling).
- **Data Collection & Analysis:**
  - ✓5-point Likert scale questionnaires.
  - ✓SPSS Analysis: Descriptive statistics, One-way ANOVA, Linear Regression.

Findings – Laboratory Conditions & Demographics

- **Level of Disorganization (Mean Perception Scores):**
  - ✓**Teachers:** Mean = 2.90 (SD = 0.4840) → *Moderate Disorganization*
  - ✓**Students:** Mean = 3.24 (SD = 0.5879) → *Moderate Disorganization*
- **Demographic Independence (ANOVA):**
  - ✓Teacher Age (p = 0.342, H = 2) → No significant effect.
  - ✓Length of Service (p = 0.414, H = 2) → No significant effect.

•*Key Takeaway:* Disorganization is systemic across schools, regardless of teacher experience.

Findings – Perception Gap on Environmental Impact

- **Teacher Linear Regression Model:**
  - ✓R^2 = 0.925 (F(1, 43) = 161.007, p < 0.001, beta = 0.962)
  - ✓Strong association between disorganization and perceived negative environmental/safety impact.
- **Student Linear Regression Model:**
  - ✓R^2 = 0.003 (F(1, 284) = 0.756, p < 0.001, beta = 0.056)
  - ✓Negligible variance explained; very weak perceived association.
  - ✓*Interpretation:* Highlights a critical awareness gap—students fail to perceive cumulative or hidden chemical hazards.

Discussion – Safety Gaps & Governance

- **Critical Operational Gaps:**
  - ✓Routine absence of basic Personal Protective Equipment (PPE) like goggles and gloves.
  - ✓Lack of chemical waste management (direct discharge into open environments).
  - ✓Accumulation of expired, hazardous reagents due to a lack of disposal infrastructure.
- **Educational Consequences:**
  - ✓Transition from hands-on practical inquiry to theoretical rote learning.
  - ✓Discomfort, headaches, and eye irritation reported during lab work due to inadequate ventilation.

Proposed Complementary Digital Learning Model

- **Core Digital Framework Components:**
  - ✓**Virtual Laboratories & Simulations:** Practice hazardous, toxic, or high-cost reactions safely.
  - ✓**AI-Supported Assistants (e.g., ChatGPT, Rasa):** Provide immediate feedback and step-by-step safety guidance.
  - ✓**3D Molecular Visualization:** Aids understanding of complex bonding and micro-structures.
- **Pedagogical Benefit:**
  - ✓Allows unlimited repetition of experiments without reagent consumption or chemical waste.

Implementation Reality – The "Digital / AI Divide"

- **Infrastructure Challenges in Jaffna:**
  - ✓ Unreliable power supply and limited internet connectivity.
  - ✓ Functional computer lab shortages in resource-constrained schools.
- **Human Factor Barriers:**
  - ✓ Teacher technology-related stress and gaps in professional digital training.
  - ✓ Language accessibility for AI platforms.
- ***Equity Warning**:* Unplanned digital adoption risks widening educational disparities between well-resourced and rural schools.

The Blended Approach (Physical + Digital Integration)

- **What Digital CANNOT Replace:**
  - ✓Tactile/manipulative skills (titration, accurate weighing, glassware assembly).
  - ✓Real-world reaction observation, physical safety drills, and waste sorting.
- **Optimal Blended Strategy:**
  - ✓Use digital labs for **pre-lab preparation**, abstract concepts, and high-risk simulations.
  - ✓Retain physical labs for core hands-on competency development.

Conclusions & Future Recommendations

- **Conclusions:**
  - ✓ Lab disorganization presents serious environmental, safety, and learning challenges in Jaffna.
  - ✓ A striking gap exists between teacher concern and student awareness regarding lab risks.
- **Recommendations:**

1. Governance: Establish safe chemical waste protocols and student safety education.

2. Digital Pilot: Conduct pilot studies to measure learning achievement, usability, and cost-effectiveness.

3. Policy: Focus infrastructure investments to bridge the digital divide before full-scale adoption.

Thank you for your attention.